

Inclusive AI Fashion Stylist with Measurement and Color-Blind Safety Analysis

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Project Proposal Report

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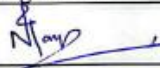
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The selection of appropriate clothing size according to body measurements is a common problem faced by many consumers while purchasing clothing. Most consumers are not aware of their body size and measurements. Therefore, they are forced to test various sizes of clothing before finding the appropriate one. Various studies have been conducted on possible solutions to this problem, including conventional size charts, virtual fitting rooms, and AI-based fashion recommendation systems. However, these systems are mostly based on general size charts and body models, which do not accurately reflect the diversity of human body shapes and sizes. In addition, some systems may be based on body type classification and fashion tips, not clothing size prediction. Moreover, some systems may be applicable to only one gender, either male or female. In addition, there are systems that predict body size, and others predict clothing size based on body weight. However, the proposed system is focused on clothing size analysis applicable to both genders. Moreover, it offers fashion tips to improve body look. In order to address these challenges, the proposed system will compare the user's information with the body measurement data collected in advance. In addition, it will use images of the user's body to predict clothing size and offer fashion tips according to body type. The main focus of the proposed system is to analyze clothing size using images of the body, particularly the torso region. In addition, it will apply image processing techniques using a grid-based measurement approach and deep learning techniques such as CNN and YOLO to detect and measure body size. The main objective of this study is to propose a system that can predict clothing size more accurately, saving time and improving customer shopping experience.

Keywords: Clothing Size Prediction, Image Processing, Fashion Technology, Fashion recommendation

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LIST OF ABBREVIATIONS

*Table 1*List Of Abbreviations

Abbreviation	Full Form
RP	Research Proposal
CNN	Convolutional Neural Network
YOLO	You Only Look Once
AI	Artificial Intelligence
TIM	Technology Integration and Management
SDG	Sustainable Development Goal
SVM	Support Vector Machines
ML	Machine Learning
FR	Functional Requirement
DL	Deep Learning

1. INTRODUCTION

1.1 Background and Context

The fashion retail industry has become more improved today, both physically and digitally, than it was in the past. Nevertheless, there still remain many challenges in the retail industry. The major challenge is that the consumers face problems, for instance, they still cannot accurately determine the exact size they need when purchasing clothes. This challenge in the physical store results in long queues in the fitting rooms. Especially during peak shopping seasons, such as the New Year season, there are long queues at the cloth stores. Some consumers need to return the clothes they purchased because they were still undetermined about the exact size when purchasing the clothes for someone else as a gift. [93% of clothing returns occur due to incorrect size or fit [1]].Moreover, there is also the challenge of the incorrect sizes in the physical store and online platform, where the consumers return the clothes they bought from the online shopping platforms because they were hard to determine the exact size.

Consumers all over the world have also experienced this challenge and are struggling with the lack of size consistency. The research indicates that the uncertainty associated with the size/fitting of apparel strongly influences consumer purchasing behavior. A McKinsey report indicated that 62% of apparel consumers have stated that uncertainty about size and fitting is a key obstacle to purchasing apparel, and 38% of shoppers have left a location without making a purchase because the fitting rooms were crowded [2]. Though both genders have a problem with this issue, women tend to be impacted more than men due to the variance in women's body types and shapes compared to men's, as well as the greater number of apparel designs and styles within women's apparel than men's apparel. Studies show 64% of consumers cited "poor fit" and/or "fit issue" as their primary reason for returning apparel items, and 62% of consumers claim to have a hard time finding appropriately fitting apparel at retail stores [3] .

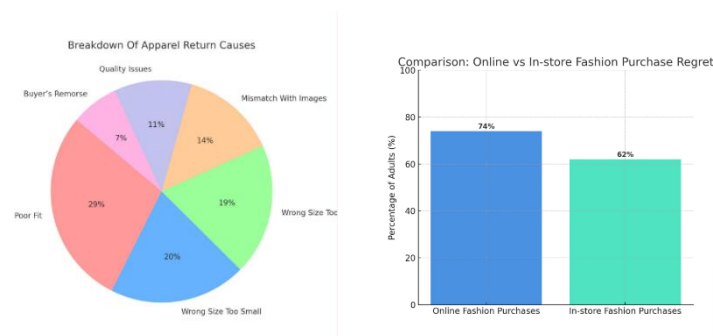


Figure 1.1 colorsocks return article

Customers are frequently assisted in selecting clothing sizes by current methods including manual body measurements and conventional size tables. Virtual fitting rooms and AI-based size recommendation systems are examples of current solutions, however many of them rely on little datasets or generic body models, which can lower the accuracy of size predictions. As a result, advanced systems that offer dependable and customized sizing suggestions are required.

The proposed system is to create an AI-Based Virtual Intelligent Size Recommendation system using image processing and grid-based measurement techniques to identify the user's size and suggest the best size to the user based on the Recommended Clothing Size for Garments (T-shirts, blouses, and frocks). Also provide the user with suggestions that help enhance body appearance, allowing users to get better body shape

The suggested solution seeks to reduce clothing size selection uncertainty and improve productivity in both physical and virtual retail settings by combining artificial intelligence and computer vision techniques.

In general, the present study aligns with Sustainable Development Goal 8: Decent Work and Economic Growth. By lowering fitting room traffic, product return rates, and customer happiness, it is suggested that the use of AI-based size recommendation technology can improve productivity and efficiency in the fashion retail industry. The suggested technologies will support sustainable economic growth in the fashion retail industry by encouraging smart digital retail practices.

This research is also in line with the Technology Integration and Management (TIM) research cluster. This is because the proposed research will integrate emerging technologies such as artificial intelligence, computer vision, and data-driven decision systems into retail management practices. By combining technological innovation with retail problems, the proposed solution will enhance customer experience while promoting efficient retail operation in the fashion retail sector.

1.2 Literature Review

The problem of clothing sizing has become an important research area in the fashion technology and e-commerce industries. Researches has explored several approaches to improve clothing size recommendations, including machine learning, computer vision, body measurement estimation, and virtual trying-on systems.

Numerous recent studies have explored using a variety of machine learning models to help identify appropriate clothing sizes based on the user's anthropometric measurements. One such study looked at developing predictive models to determine clothing sizing and used models like Support Vector Machines (SVM) together with 3D data collected from a system that can capture body shape information using a variety of techniques. In conclusion, the limitation here is researchers have taken a small dataset. However, there is size and age distribution imbalance [4].

Another area of existing research is AI-based body type analysis systems that recommend clothing that is appropriate for body types. One study proposes a machine learning-based fashion recommendation system that uses Convolutional Neural Networks (CNN) to analyze the body types of male users and generate personalized fashion suggestions. The body type classification system had great performance overall; however, the scope of the body type classification was specific to classify body types, not to predict clothing sizes. Therefore, this type of system would not provide adequate recommendations when users want specific clothing sizes. Additionally, since it was designed only for men. Although the research not tested with many different ethnic groups and also not tested mobile or edge devices [5].

Creating virtual dressing-room systems which provide users an opportunity to see how various types of clothes would fit them based on photographs taken from different perspectives. These systems utilize both computer-generated camera models along with algorithms that determine how well garments should fit based on size. However, many virtual try-on systems rely on simplified body models or predefined avatars that do not accurately reflect real human body measurements, which can lead to inaccurate size recommendations. Furthermore, it often focuses on how the clothes look when worn and recommends items that fit. It does not show the exact size of the clothes [6].

The recent studies adopt 2D-3D non-contact anthropometric method for daily dressing state to estimate body measurements and online clothing recommendation. The methodology adopted in this study is to collect images and data of bodies, analyze body shapes by using curves, and

create mathematical function for it. The limitation of this study is the study only considers young Asian women, so it may not be effective for men, older people, or different ethnicities. The methodology mainly focuses on body measurement estimation but does not consider automatic clothing size recommendation or prediction [7].

Three cities in Ghana were selected for the study: Acura, Kumasi, and Tamale. The study targeted weight control and its effect on body image for both fashion models and non-models. The instruments used in the study were a questionnaire, weighing scales, and stadiums. The limitations in the study were the use of collected body measurements through the use of weighing scales and tape measures. It only showed the effect of body measurements and weight control on body image for both models and non-models. It also has a small dataset since only 192 participants were targeted. Questionnaires are also based on the honesty of the participants, and this may cause bias in the results [8].

This research examines the effect of clothing size on the self-esteem of men and women. In this article, the research focuses on conducting a literature review of previously published research and analysis of that data. Limitations of the study include a lack of original data, and the limitations placed on participant age groups (only young adults will be analyzed) [9].

The research of this study proposes how to predict the measurements of a young woman from the 3D body scans. This study will analyze 220 females by using the [Victor Smart LC3] 3D Body Scanning system and clustering algorithms based on the body shape classification. The study uses a data-driven quantitative approach. The majority of women are within the age range of 18-25; thus, only young, Chinese women are being measured with this study [10].

1.3 Research Gap

Existing RP use different technologies such as machine learning (SVM), body scanning, cluster analysis, and virtual fitting systems to estimate the size of clothing. Some research has focused separately men or women or by age group. Similarly existing research faces various limitations. Further, the current solutions have limited ability to provide the diversity of human shapes and measurements, because they mostly use standardised size charts, standardised shape models, or limited anthropometric databases. Furthermore, much of the research has mostly focused on classifying body type and providing visualisation of virtual garments, rather than providing accurate, personalised size recommendations for individuals. Many require specific hardware (such as 3D body scanners) or a controlled space to operate, making them impractical for everyday users to use in retail situations.

Table 2 research gap table

Features	1	2	3	4	5	7	Proposed Solution
Limited for gender	✗ both	✓ Male	✗ both	✓ Female	✓ Female	✓ Female	✗ both
Limited age gap	✓ Young adults	✗	✗	✓ 18-26	✓ 18-29	✓ 18-25	✓ Above 15
Automated measurement	✓	✓	✓	✓	✗	✓	✓
All body sizes	✗ most 'M'	✓	✓	✓	✗	✓	✓
Large Dataset	✗	✓	✓	✗	✗	✗	✓
Provide fashion suggest	✗	✗	✗	✗	✓	✗	✓
Predicts Body size (not type)	✗	✓	✗	✓	✗	✓	✓

By using combination of image processing and grid base measurements, this AI powered virtual intelligence size recommendation system can accomplish tasks that were previously impossible, eliminating all of the above challenges. The proposed system is different from other existing systems in terms of health tips and weight loss recommendations after analyzing body images. Some of the existing systems are capable of estimating body sizes using uploaded images, but they are mostly focused on body analysis rather than providing accurate clothing sizes. The proposed new method would not only provide a clothing size recommendation but would also provide the user with suggestions that help enhance body appearance, allowing users to get better body shape.

2. OBJECTIVE

2.1 Main Objective

To develop a new AI-based virtual intelligent size recommendation system that uses image processing, grid-based methods, and deep learning methods (CNN, YOLO) to accurately predict clothing size for users based on their torso images and body measurements, such as for T-shirts, blouses, and dresses, for both males and females, and provide fashion tips to enhance the look of the user's body.

Objective Explanation

This study suggests an AI-based Virtual Intelligent Size Recommendation system made specifically for clothing items including T-shirts, blouses, and dresses for both in order to overcome these limitations. The approach focuses on examining the human torso, which is upper part. Participants will first submit body measurements and photos using a safe server-based data collecting procedure after receiving the appropriate ethical clearance. A reference database of body measurements and size trends will be built with this dataset. When a new user accesses the software, the system uses mathematical measurement models/grid base image measurement method and image processing techniques to assess the user's dimensions and body image. It then compares the results with the previously gathered dataset to determine the appropriate clothing size. Also identifies a size which may be more suitable for achieving a proper fit, in addition to providing guidance on how to achieve it [fashion tips]. In order to solve the gaps in the previous studies, a more comprehensive system is proposed. Most of the previous studies are focused on offering health tips or fashion tips based on body weight [11], and a few studies have been focused on estimating clothing sizes by using images. The proposed system integrates both of these ideas by first predicting the clothing sizes of the user and then offering fashion tips to enhance body appearance based on the predicted sizes.

2.2 Specific Objectives

- Building an inclusive dataset of user details/ body images and measurements through a data collection method verified by ethics approval.
- Developing an image processing technique to identify and extract the torso region of user images.

- Develop a grid-based mathematical model to calculate user body proportions and clothing sizes.
- Creating and train an AI based recommendation system to use comparative analysis between new user data and previously stored measurements
- Creating a user-friendly interface where users can upload photos of themselves and be provided with fashion recommendations and sizes that would improve the shape of their bodies and give tips.

3. METHODOLOGY

- Research Approach

The research approach that will be utilized in designing and evaluating an image-based clothing size prediction system is an experimental research approach. This research approach will be useful in designing an image-based clothing size prediction system, where the research will be focused on designing an image-based clothing size prediction system that can analyze images of users' bodies and provide recommendations on clothing sizes. This research approach will be useful in designing an image-based clothing size prediction system, where the research will be focused on designing an image-based clothing size prediction system that can analyze images of users' bodies and provide recommendations on clothing sizes and provide fashion tips.

- Technical Methods and Algorithms

Object detection algorithms, such as YOLO (You Only Look Once), will be utilized in detecting the user's body regions from images. In YOLO remove unnecessary background. This algorithm will be useful in detecting the torso region, including the chest and upper body regions. Once the torso region has been detected, the grid-based measurement method can be utilized in estimating the user's body proportions, including the width of the chest and the length of the torso. Convolutional Neural Network algorithms can be utilized in analyzing image features and determining relationships between user body proportions and clothing sizes.

- Tools and Platforms with Justification

The proposed system will utilize image processing and machine learning technologies to analyze user images and estimate clothing sizes. The proposed system will be developed using Python programming to train and develop the image-based clothing size prediction system. The use of digital images to analyze user information will allow the system to efficiently process user body information while enabling scalability for potential mobile or web-based system development.

- Data Collection and Ethical Issues

The data will be collected using the survey method after obtaining ethical clearance. Participants will be asked to voluntarily participate in the study and provide their body

measurements and images. The data collected will be kept confidential and will be used for research purposes only.

- Validation Strategy and Performance Metrics

The system will be validated by comparing the clothing size predicted by the system and the actual clothing size of the participant. Accuracy, precision, and recall will be the performance metrics of the system.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

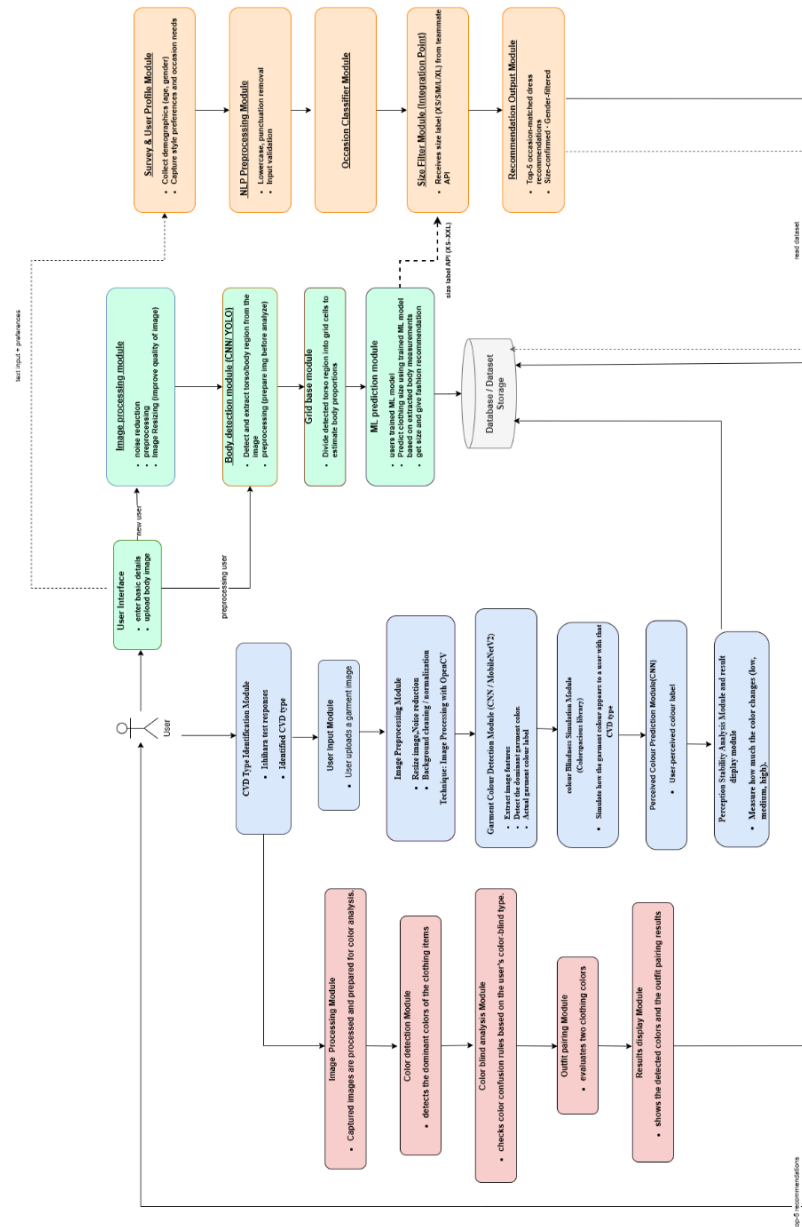


Figure 4.1 High level system diagram

The high-level system architecture provides a general view of the overall workflow of the proposed AI-based clothing size recommendation system. First, the user will input the basic information and upload the body image through the user interface. Then, the image processing module and body detection module will analyze the body image through CNN or YOLO algorithms. Finally, the grid-based measurement module and ML prediction module will analyze the measurement data and provide the clothing size recommendations along with fashion recommendations.

5. User Requirement

5.1 Stakeholder Identification

There will be numerous parties that are impacted by the proposed system or will interact with the systems.

- Main Users (Customers) - Would like to determine their exact body size and measurements for clothing items (such as blouses, t-shirts & dresses) by uploading images and measurements to the system.
- The System Administrator - Is responsible for overseeing the management of the database, monitoring the use of data by users and ensuring that the system is functioning properly.
- Researchers/Developers - Individuals working on the development, enhancement and evaluation of an AI-based size recommendation model.
- Fashion Retailers (Indirect Stakeholder) - Clothing retailers who will see benefits from this system in regard to reducing overcrowding within fitting rooms while improving customer shopping experiences.

5.2 Functional Requirement

FR1 – The system allow the user to input their basic details and body measurements through the data collection form.

FR2 – The system allow the user to upload images for analysis of body measurements.

FR3 – The system should be able to detect the torso area from the uploaded image using image processing algorithms.

FR4 – The system aims to apply grid measurement algorithms to determine body measurements such as chest and torso length.

FR5 – The system enable use machine learning and CNN algorithms to analyze the features extracted from the body and determine the appropriate clothing size.

FR6 – The system compare the input data for the user with the measurements collected from other users and input into the dataset to improve the accuracy of body size determination.

FR7 – The system determine the appropriate clothing sizes for clothing such as T-shirts, blouses, and frocks.

FR8 – The system provide fashion ideas to help the user improve their appearance, such as losing weight tips.

FR9 – The system allow the administrator to manage the user dataset used for system analysis.

5.3 Non Functional Requirement

- security
 - NFR1 requires a secure method of storing user body image files and body measurements in a secure database. Access to these images will be restricted to only those administrators who are authorized to access the database.
 - NFR2 states that user body images will only be used for research purposes or research-related purposes, and they may not be shared with anyone outside of the user's consent.
- Performance
 - NFR3 requires that uploaded body image files are processed within a short time frame once the image has been uploaded in an acceptable time frame.
 - NFR4 requires that the system be capable of allowing many users to access the system at the same time, without any negative effect on the system's performance.
- Reliability
 - NFR5 – The system shall maintain consistent accuracy in predicting sizes after repeated use.
 - NFR6 – The system shall maintain a minimum system availability of 95% during system hours of operation.
- Privacy
 - NFR7 – The system shall obtain consent from users before obtaining body images and measurements using an informed consent form.
 - NFR8 – The system shall treat participant data in a confidential manner according to ethical research practices.

6. COMMERCIALIZATION PLAN

- Target market and customer persona –

The primary target market will comprise individual consumers who are above the age of 15. This will include males as well as females the clothing such as T-shirts, blouses, frocks. The consumers will face a problem in finding their exact clothing sizes while shopping from stores or websites. This will particularly affect young adults who shop a lot. The secondary target market will comprise fashion retail companies that can incorporate the system in their mobile applications or websites.

- Value proposition

The proposed system will enable users to receive accurate size recommendations based on their images. This will enable users to know their correct sizes without the need to try on different sizes. The system will also save time for users since they don't have to wait for their turns to try on clothes. The system will also offer users fashion tips and body improvement ideas.

- Revenue model

The proposed system can be monetized in several ways, and one of these ways is a software-as-a-service model, where fashion retailers will be able to subscribe to our service and integrate our system into their website or mobile application.

Another way is a mobile app freemium model, where users will be able to download our mobile app for free, with limited usage, and then be able to access more features, such as detailed size analysis, with a premium subscription.

- Cost estimation

The key cost components would be cloud storage or hosting for storing images and data uploaded by users, as well as server cost for running machine learning models. Extra cost components would be for maintenance, updating, and security of the system.

- Pricing strategy –

System will allow dual pricing strategy since they are two target markets individual users and fashion retails. Individual users can access basic size recommendation for free with limited usage, while advanced features are available through a medium-cost premium subscription. Fashion retail companies can use the system through a subscription or API integration by paying a monthly fee. This strategy helps to reach a wider market area.

- Competitive advantage

current solutions are based on traditional size charts, virtual fitting rooms and recommendation tools with limited accuracy. The proposed system improves this by using image processing, grid-based measurements increase accuracy in size prediction. The system also offers users personalized fashion tips, which is an added value beyond just size prediction.

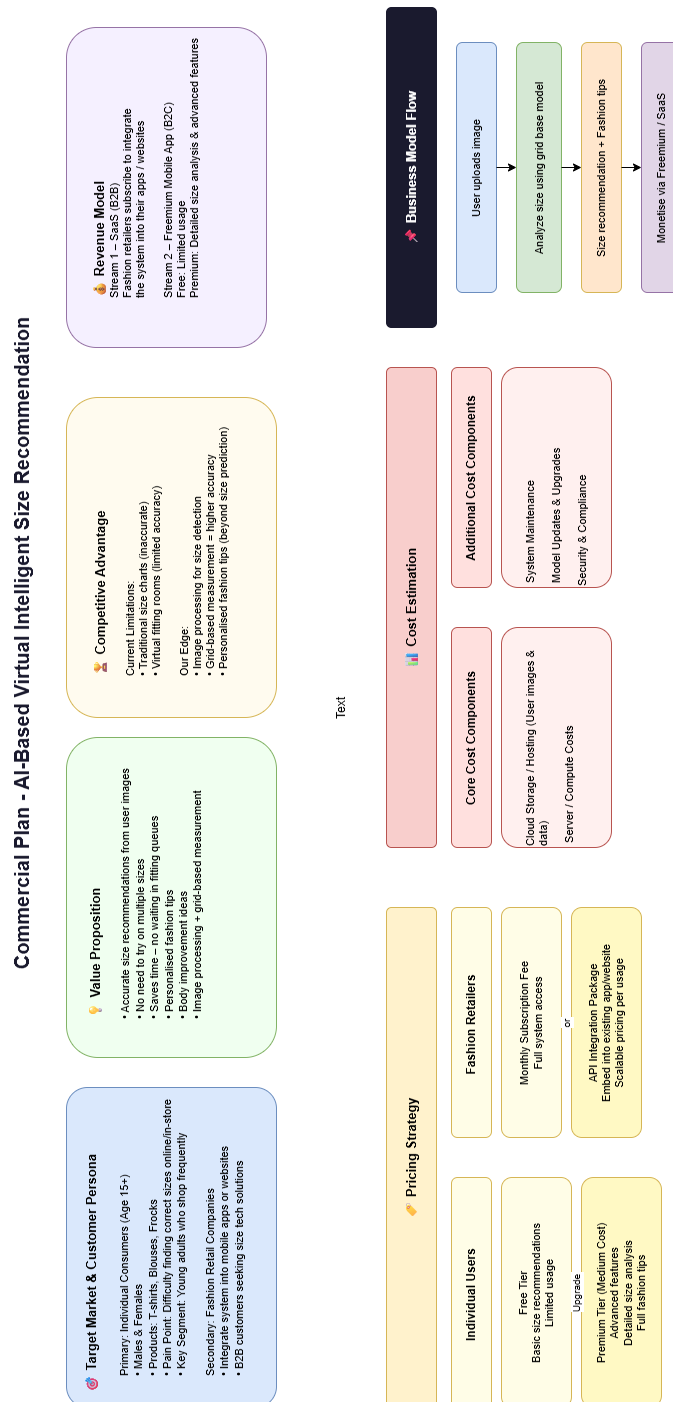


Figure 6.1 commercial plan

7. BUDGET AND JUSTIFICATION

Table 3 budget

Technology /Resource	Total
Software and Services	
Cloud Storage and Hosting (Firebase/AWS – 10 months) – to host the trained model and store system data for online access	55,000.00
Wi-Fi / broadband connection (10 months)	25,000.00
Data Collection	
Data Collection Materials, Testing & Documentation	10,000.00
Contingency & Miscellaneous	
Unexpected Expenses (10% buffer)	9000.00
Total	99,000.00

8. WORK BREAKDOWN STRUCTURE (WBS)

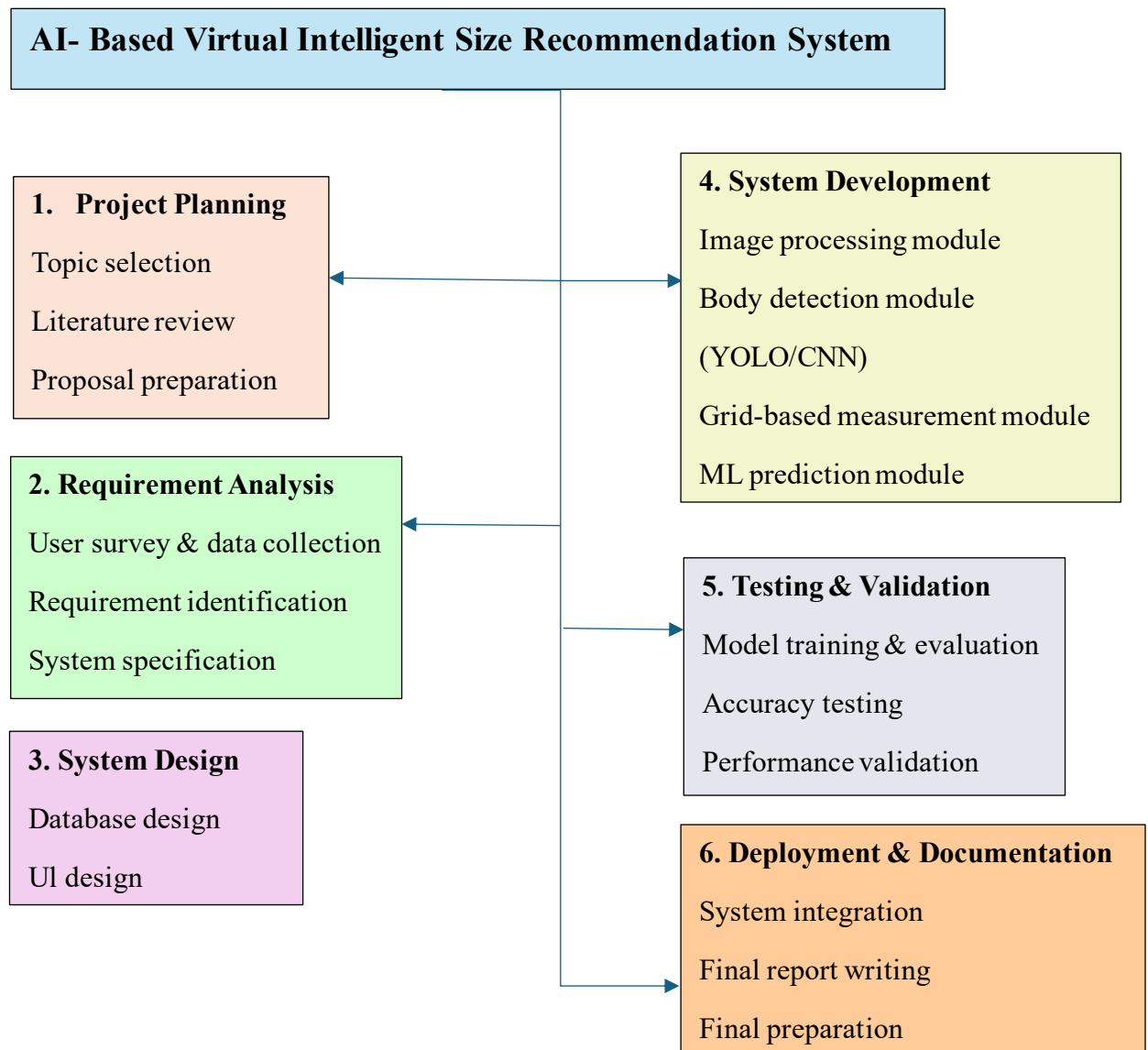


Figure 8.1 WBS

The Work Breakdown Structure (WBS) describes the major components of creating the AI-Based Virtual Intelligent Size Recommendation component. The components were conducting a requirement analysis, obtaining data, designing a system architecture, creating an image processing system and creating a machine learning model. These components will be completed in three phases: building the AI-Based Virtual Intelligent Size Recommendation System; building the system; and testing the system and validating the system and finally documenting the entire process.

9. GANTT CHART

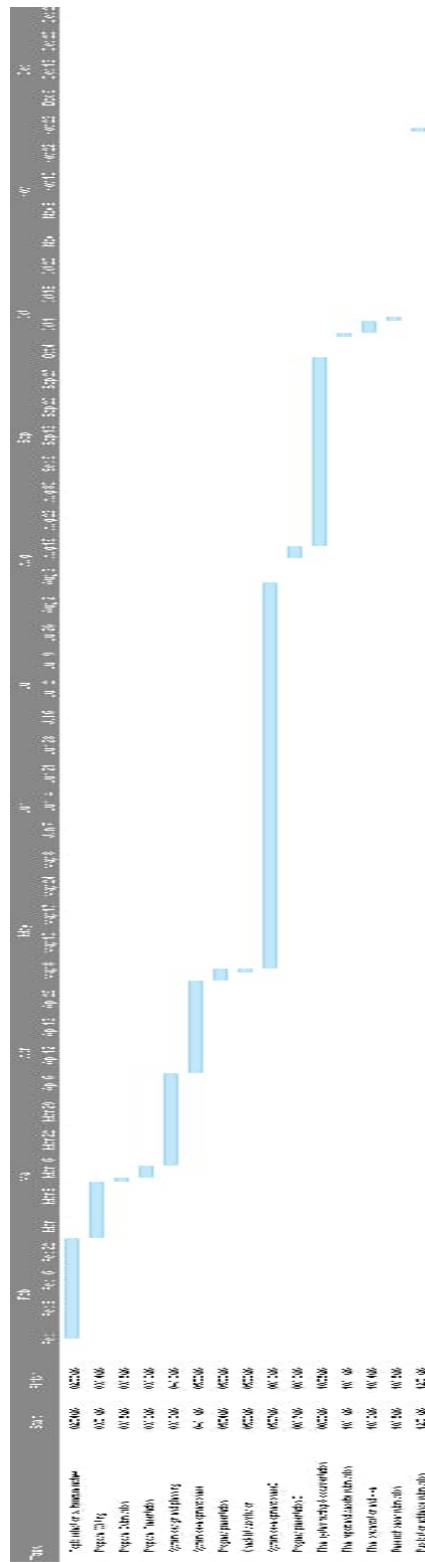


Figure 9.1 Gant chart

<https://drive.google.com/drive/folders/13MkU9CB6YNftDfOb360aemtZQGRHKxTc?dmr=1&ec=wgcd-drive-%5Bmodule%5D-goto>

REFERENCES LIST

- [1] P. Dovgopol, "LinkedIn," 25 December 2025. [Online]. Available: <https://www.linkedin.com/pulse/problem-size-selection-online-clothing-stores-why-returns-dovgopol-ciiif?utm>.
- [2] G. Holden, "Alibaba.com," 2024 . [Online]. Available: <https://www.alibaba.com/product-insights/ai-fashion-try-on-apps-vs-in-store-fitting-rooms-do-virtual-models-reduce-return-rates-meaningfully.html?utm>.
- [3] "Colorful socks," 2025. [Online]. Available: <https://bestcolorfulsocks.com/blogs/news/clothing-fit-regret-statistics?utm>.
- [4] RuqeyAlhassawi, SimeonGill, Steven Hayes and Kristina Brubacher, "Evaluating machine learning Evaluating machine learning prediction using anthropometric measurements from 3D body scanning," 2019.
- [5] Sallar Sami Jogi, Bushra Khan, Safiullah jamali, Zafar Ali, Atta ur Rehman, Naseebullah and Shahzaib Ali, "AI-powered Body Type Analysis for Fashion Recommendation System," 2025.
- [6] Mrs. Sindhu Jain A M, Anusha K, Tanushree N, Thushara K and Varalakshmi D R, "Online Clothing Recommendations and Virtual Trials using AI," 2025-03-08.
- [7] Ke Ke Sun, Yi Jui Chiu, Yi Cheng Chen and Yu Xiu Wang, "A 2D-3D non-contact anthropometric method for daily dressing state-takes young Asian women as example," 2017.
- [8] Benard Edem Dzramedo and Tahiru Alhassan, "Fashion Models and Body Image: A Comparative Study on the Effects of Weight Controls and Feeding Behaviours of Models and Non-Models," 2024.
- [9] C. Mahoney, "The Effects of Clothing Size on Self-Esteem regarding Gender," 2022.
- [10] zhengtang tan, shuang Lin and zebin wang, "Cluster Size Intelligence Prediction System for Young Women's Clothing Using 3D Body Scan Data," 2024.
- [11] Obes Metab Syndr and Ju Young Kim , "Optimal Diet Strategies for Weight Loss and Weight Loss Maintenance," 2020.

10.APPENDICES

- Survey instruments

A survey was conducted to gather information from people about their experiences and problems when it comes to choosing the right clothing sizes while shopping.

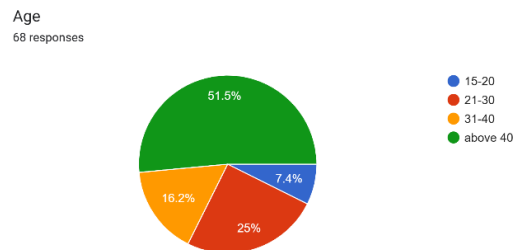


Figure 10.1 Age chart

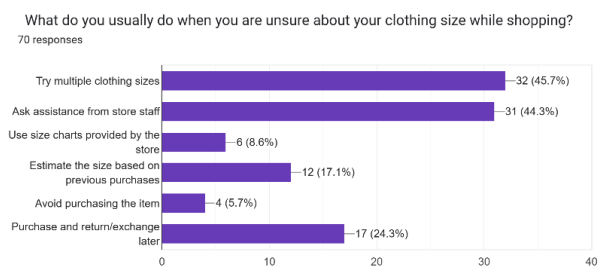


Figure 10.2 Unsure Clothes size chart

How useful do you think an AI-based system that recommends clothing sizes using body images would be
69 responses

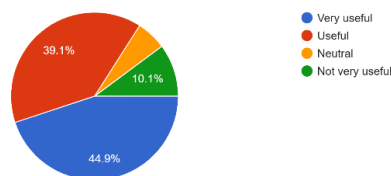


Figure 10.3 AI system recommendation chart

<https://docs.google.com/forms/d/1qb9MmOQUqhlPR772LT3vakF2d8C24qC3sjofh9Zbkh/edit#responses>

<https://docs.google.com/spreadsheets/d/1T1T9BuQiin-1PRFsnydAI0v4mP3SqZX-39ohixNMGaY/edit?resourcekey=&gid=1051557128#gid=1051557128>

- Detailed diagrams

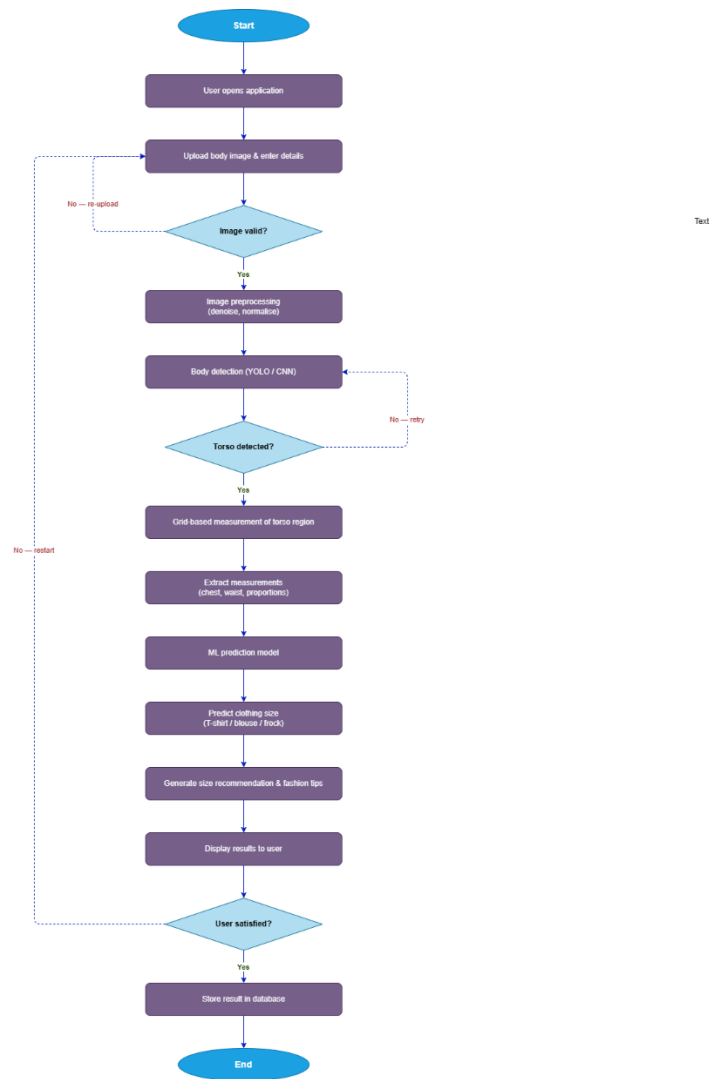


Figure 10. 4 Detail diagram

- Risk analysis

Table 4 Risk Table

Risk	Impact	Mitigation Strategy
Insufficient dataset for training	Low model accuracy and unreliable size predictions	Collect more user data through surveys and image submissions and use data augmentation techniques
Difficulty in detecting body regions accurately	Difficulty in detecting body regions accurately	Provide image capture guidelines and apply image preprocessing techniques
Users may hesitate to share body images	Users may hesitate to share body images	Obtain ethical clearance and ensure secure data storage with user consent
Incorrect clothing size recommendations	Incorrect clothing size recommendations	Continuously improve the model using additional training data and validation testing
Slow processing when many users access the system	Slow processing when many users access the system	Use scalable cloud infrastructure to handle higher processing loads